



Synthesis of Phenyl-o-Thiourea and Evaluation of its Anticorrosive Property in Alkaline Media on Corrosion of Aluminium

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Received: 01.03.2026 Accepted: 17.04.2026 Published: 30.06.2026ss

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ABSTRACT

This work explores the synthesis of a Phenyl-o-thiourea and its application as an organic corrosion inhibitor to the corrosion of aluminium in alkali solution. In the synthesis of inhibitor thiocarbamylation was performed and the compound was coupled with phenol and the inhibitory behaviour of the compound was studied in a systematic way by weight-loss, adsorption modelling, electrochemical impedance spectroscopy (EIS) and potentiodynamic polarization. The experimental results from the weight-loss measurement demonstrated that the inhibitor is highly protective in lowering dissolution of aluminium in 1 N NaOH and the rate of inhibition increased steadily with concentration and temperature. Phenyl-o-thiourea adsorption on the aluminium surface was observed to obey the Langmuir adsorption isotherm and the thermodynamic parameters calculated in the calculation demonstrated that mostly a physisorption-based process was followed. The results of EIS analysis indicated that there was an increase in charge-transfer resistance and consequent fall in double-layer capacitance which was a sign of a protective adsorbed film. The experiment of polarization determined that the compound is a mixed type-inhibitor in that both cathodic and anodic reactions were inhibited. In general, Phenyl-o-thiourea has a high potential of being used as an effective corrosion inhibitor of alkaline environment on aluminium to provide stable surface coverage at elevated temperatures.

Keywords: Phenyl-o-thiourea:Aluminium corrosion; Corrosion inhibition; Electrochemical impedance spectroscopy; Potentiodynamic polarization study.

1. INTRODUCTION

The great strength to weight ratio, good formability and desirable conductivity make aluminium very applicable in structural, transportation, electrical applications and packaging as well. Although aluminium is generally regarded as a corrosion-resistant metal, its corrosion resistance strongly depends on the presence of a stable, adherent and self-healing oxide film on the surface (Hatch *et al.* 1984). This protective Al₂O₃ film ensures passivity under many environmental conditions, particularly in neutral and mildly acidic media. However, aluminium behaves quite differently in alkaline environments. In alkaline solutions such as sodium hydroxide (NaOH), the passive film is chemically attacked and dissolved, leading to rapid and

aggressive corrosion (Sherif *et al.* 2006) (Ebenso *et al.*, 2007). This dissolution produces aluminate ions, exposing fresh metal surface and accelerating the corrosion cycle (Rayung, *et al.* 2020; Gece *et al.*, 2008). Such behavior poses critical challenges in industries where aluminium is exposed to alkaline cleaners, surface-treatment solutions, and waste-alkali streams. Therefore, controlling the corrosion of aluminium in NaOH becomes an important scientific and technological issue.

One of the promising approaches for mitigating corrosion in aggressive environments involves the use of inhibitors. The processes of inhibitions noted by corrosion inhibitors include adsorption of the molecules onto the metal surface, decreases the active sites of

corrosion, and inhibits the rate of anodic, cathodic or mixed reactions ((Bharatiya *et al.* 2019; Ammar *et al.* 2008). Organic inhibitors are particularly attractive because many contain heteroatoms such as nitrogen, oxygen or sulphur, along with π -electron systems that can strongly adsorb on metal surfaces (Ebenso *et al.* 2007; Jenkins *et al.* 2009). In alkaline media, adsorption of these molecules can create a barrier that slows dissolution of the passive film and suppresses metal attack. Green inhibitors derived from plants contain polyphenols, alkaloids and flavonoids, which exhibit strong adsorption and are environmentally safer than synthetic chemicals (Raja *et al.* 2008; Liu *et al.*, 2014).

The knowledge of the inhibition mechanism would be critical in developing effective and sustainable corrosion-control policies. Most inhibition mechanisms are studied using adsorption isotherms, potentiodynamic polarization and electrochemical impedance spectroscopy (EIS), which provide information on corrosion current, charge-transfer resistance and film-formation behavior (Mansfield *et al.* 1990; Mahdi *et al.*, 2024). In recent years, computational chemistry and quantum-chemical descriptors have further improved the understanding of adsorption energies and corrosion-inhibition potentials, which help to predict new and effective inhibitor molecules (Gece *et al.* 2008; Mallaiya *et al.*, 2011). The present work is aimed to synthesise and analyse corrosion inhibition efficiency of phenyl- o-thiourea on aluminium in alkaline solution.

2. EXPERIMENTS

2.1 Aluminium Specimen

Weight loss measurement was done using rectangular Aluminium specimens in form of coupons whose size was 1.5 cm $\times$ 2.4 cm. The surfaces of coupons were polished using 1/0, 2/0, 3/0 and 4/0 emery papers, and then lastly cleaned using organic solvent, acetone. A hole with a diameter of 0.5 cm was drilled close to the edge of the coupon to allow hooking it to the glass rod on which it was immersed. Weighed specimens were placed in the 1 N NaOH solution and after 2 h the specimens were taken from 1N NaOH solution, washed with distilled water, decreased with acetone, dried and weighed. The specimens of Aluminium were placed in 1N NaOH blank solution and in 1N NaOH solution with inhibitor at different concentrations for 2 h.

2.2 Synthesis of Phenyl-o-thiourea

A solution of (3.14 mol) 4,4'-diaminodiphenyl ether in deaerated water (600 ml), 600 ml of concentrated hydrochloric acid and 20 g of charcoal activated were added and heated to 50° C. The mixture was then filtered and 12.7 mol ammonium thiocyanate added and the mixture stirred at a temperature between

90° C -100° C for 24 hours. The yellow granular product started to precipitate in two hours due to the reaction. The mixture was left to cool and filtration was carried out to obtain the product. Tetrazotization of the product followed by reaction with phenol is then carried out. Figure.1 depicts the structure of inhibitor.

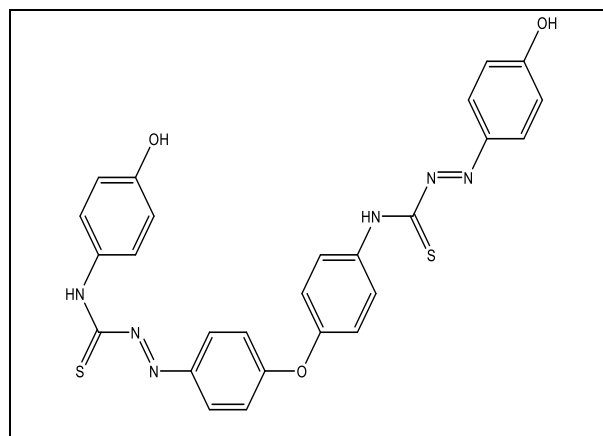


Figure.1 Structure of Phenyl-o-Thiourea

2.3 Solutions

An analytical grade NaOH pellets were used to prepare 1N NaOH solution. The preparation of NaOH solution, inhibitor solutions was done with the help of double distilled water. An inhibitor stock solution of 10mM was prepared in 1 N NaOH solution. From this stock solution the varying concentrations of the solutions of the inhibitor were diluted. Standardization of diluted solutions is done using standardized acid solutions.

2.4 Methodology

2.4.1 Weight loss method

Aluminium triplet specimens were placed in the test solution which was then immersed in 150 ml beaker containing 100 ml of 1N NaOH inhibited and uninhibited solutions. Weighing of Al triplet specimens was done before and after immersion. The weight of the specimen prior and after 2 hours of the immersion was recorded. The same method of weight loss was also used in temperature studies. The concentration of the solutions used remained constant and the temperature was changed. The experiments were conducted at 27° C, 35° C, and 40° C. This was the same procedure above that was observed. Based on the change of weight, the calculations of the percentage of inhibition efficiency (I.E%), surface coverage and the rate of corrosion were made using; (Mallaiya 2011; Mahdi 2024).

$$IE\% = \left(\frac{W_o - W_i}{W_o} \right) \times 100 \quad (1)$$

Where,

W_o = without inhibitor.

W_i = weight loss with inhibitor

$$\text{Surface coverage } (\theta) = \left(\frac{W_o - W_i}{W_o} \right) \quad (2)$$

$$\text{Corrosion rate (mpy)} = \frac{87.6 \times W}{D \times A \times T} \quad (3)$$

Where

W = weight loss

D = density in g/cc

A = area of exposure in cm^2 and

T = time in h.

The formula (Mungwari *et al.* 2025) has been used to determine the activation energy (E_a).

$$E_a \text{ (kJ/mol)} = 2.303 \times R \times T_1 \times T_2 \times \log(\rho_2 / \rho_1) / (T_2 - T_1) \quad (4)$$

Where

R = Universal gas constant

T_2, T_1 = Temperatures in K

ρ_2, ρ_1 = the rate of corrosion at temperatures T_2 , and T_1 respectively.

After determining the surface coverage (θ), the type of isotherm into which the observed experimental data fit well is determined as follows:

An inhibitor is said to obey **Langmuir isotherm** if the plot of $\log \theta / (1 - \theta)$ vs $\log C$ is linear.

An inhibitor is said to obey **Temkin isotherm** if the plot of θ vs $\log C$ is linear.

An inhibitor is said to obey **BDM isotherm** if plot of $(\log C - \log \theta / (1 - \theta))$ vs $\theta^{3/2}$ is linear.

An inhibitor is said to obey **Frumkin isotherm** if a plot of $\log (\theta / (1 - \theta)) + C$ vs θ is linear.

Where, θ = degree of coverage on the metal surface

The free energy of adsorption (ΔG_{ads}) is calculated using the formula (Mansfeld *et al.*, 1990),

$$\Delta G_{ads} = -RT \ln 55.5K \quad (5)$$

$$K = 1/(55.5) \exp(-\Delta G_{ads}/RT) \quad (6)$$

Where, K = Adsorption Equilibrium constant.

2.4.3 Electrochemical Impedance Spectroscopy

Among the many A.C methods, the method to faradaic impedances is most popular since the main advantage is that the double layer capacitance and transfer resistance of charges can be determined as a function of frequency, as the sinusoidal potential can be perturbed with an extremely small magnitude (as small as ~10 mV). Measurement can be done on the impedance of the corroding system at different frequencies through Zahner Zennium XC, an advanced Electrochemical workstation.

From R_t , efficiency is determined by the following equation:

$$IE\% = \frac{(R'_{ct} - R_{ct})}{R'_{ct}} \times 100 \quad (7)$$

Where

R_{ct} is charge transfer resistance in the absence of inhibitor.

(R'_{ct} is charge transfer resistance in the presence of an inhibitor)

The frequency where the imaginary part of the impedance is maximal ($-Z_{(ima)}$) value is calculated and the C_{dl} values are calculated by use of the equation:

$$f = \frac{1}{2 \pi C_{dl} R_{ct}} \quad (8)$$

2.4.4 POTENTIODYNAMIC POLARIZATION TECHNIQUE

The polarization studies and weight loss studies were done using the same electrode of aluminium in their composition. In the test solution, a Teflon coated Al rod with an exposed area of 0.2486 cm^2 . The electrode made out of Aluminium was then washed using the acid, distilled water and dried. The initial polishing step of the electrode was successful polishing with Emery papers 1/0, 2/0, 3/0 and 4/0. It was then washed with distilled water and degreased with acetone. Standard polarization test cells were used to conducted the polarization studies. It was made of a four hole Teflon and a glass beaker that was fully covered. The Aluminium electrode was fitted in the largest hole. The electrochemical studies were carried out using the Zahner Zennium Electrochemical work station. 100 ml of 1N

NaOH solution was used in the standard polarization test cell. Teflon coated Al rod as the working electrode, Pt electrode as the auxiliary electrode and Saturated Calomel Electrode as reference electrode, the polarization studies were carried out. Polarization curves were recorded at scan rate of 1.67mV/s that were potentiodynamic in nature. Polarization curve was traced at -200mV cathodic to the open circuit potential and proceeded in the anodic direction E_{corr} to +200mV. In the same manner the polarization curves were recorded for Al in 1N NaOH solutions with different inhibitor concentrations.

3. RESULTS AND DISCUSSION

3.1 Weight Loss Measurement

The weight loss procedure has been employed in establishing the inhibiting effectiveness and

corrosion rate of Al with different concentrations of the inhibitors and the results are presented in Table.1. The weight loss method has been performed as follows in 1N NaOH:

1. Through change of concentration of the inhibitors at room temperature
2. With changing temperature (308 k-323 k) at optimum inhibitor concentration

The efficiency of Phenyl-o-thiourea inhibitor with various concentrations is estimated using the experimental results are presented in Table.1. As indicated in the data on this table, the higher the concentration of the inhibitor, the higher the efficiency of the inhibitor as well as the slower the rate of corrosion. This implies that the inhibition process is achieved by adsorbing inhibitors on metal surface.

Inhibitor concentration (mM)	Weight loss (g)	Corrosion rate (mmpy)	Inhibitor efficiency (%)	Surface coverage (θ)
Blank	0.0796	0.4449	—	—
0.75	0.0585	0.3270	26.50	0.265
1.0	0.0444	0.2482	44.22	0.4422
1.5	0.0411	0.2291	48.49	0.4849
3.0	0.0390	0.2180	51.00	0.5100
5.0	0.0342	0.1911	57.03	0.5703
6.0	0.0284	0.1587	64.32	0.6432

Table.1 Data of Phenyl-o-thiourea in 1N NaOH at different concentrations with regard to weight loss.

One may note by examining literature on the work conducted by the early researchers that adsorption of organic substances with aromatic rings occurs by the flat orientation of the whole molecule over the metal surface due to the following reasons:

*The p electrons of the aromatic sextet interact with the unappreciated valencies of the metal atoms on the metal surface to some extent.

*The P electrons of the group, C=C, C=N, C=O, N=N,

S=O etc. also fill up the unpaired valency of the metal atoms.

*The metal atoms also react with the unpaired electrons of the atoms such as N, O, S, P etc. (Netzer 1992; Liu 2014; Zaera 1995; Jenkins 2009; Yang 2010; Pan 2008). In the existing studies the inhibitor molecule is the Phenyl-o-thiourea whose structure comprises of four aromatic rings other than N=N and C=S. Every one of these molecules of an inhibitor is consequently adsorbed on the Al surface in the flat position of entire molecules as illustrated below.

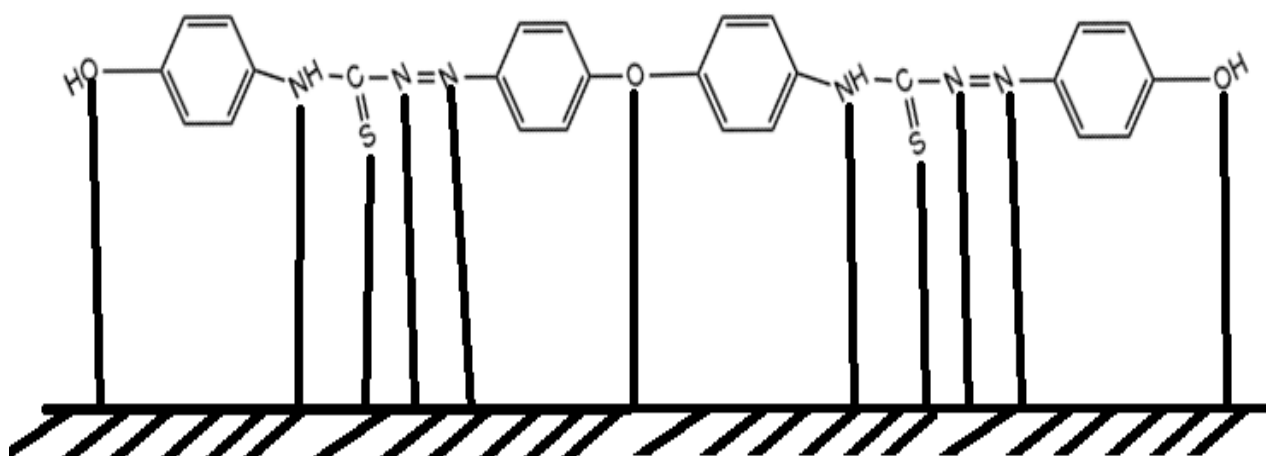


Figure.2 Adsorption of Phenyl-o-thiourea on Al surface

Organic molecules prevent corrosion since they are attached to the metal-solution interface. The adsorption is also influenced by a number of factors which include chemical structure of the inhibitor, chemical composition of the solution, nature of the metal surface, temperature and electrochemical potential at metal-solution interface. The experiment of adsorption gives the data concerning not only the adsorbed molecules but the interplay between the molecules and the metal surface.

Different adsorption isotherms have been put to test of the current experimental data and it has been established that the data is directly proportional to Langmuir adsorption isotherm. It is expressed as (Mansfeld *et al.*, 1990)

$$C / \theta = C + 1 / K \quad (9)$$

A graph of C / θ vs C must be a straight line. This was also retrieved in the experimental data and the plot is presented in Figure.3.

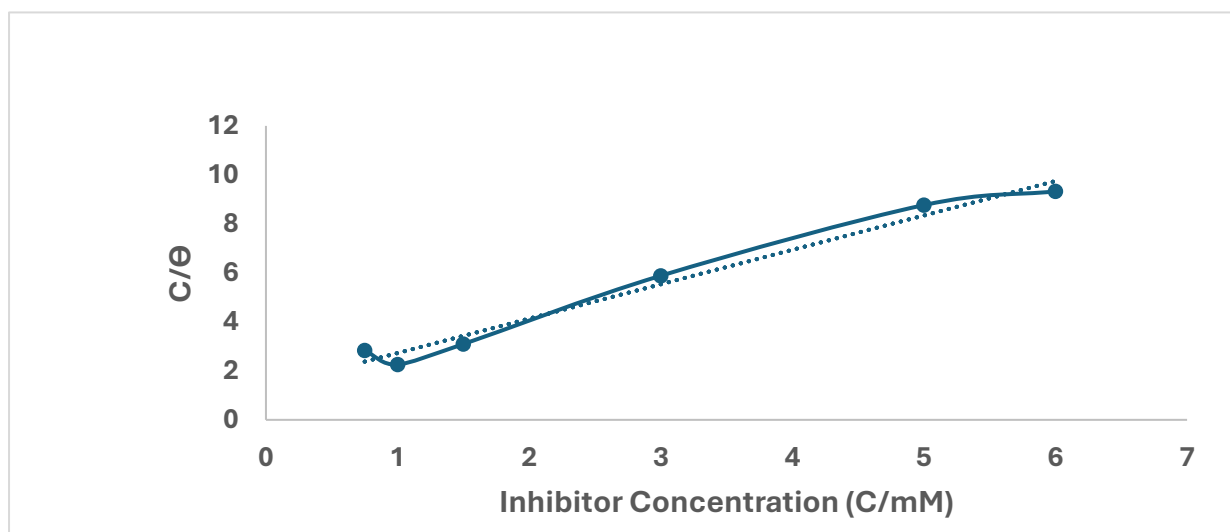


Figure.3 Langmuir Adsorption isotherm of Al corrosion under Phenyl-o-thiourea 1 N NaOH solution.

Table.2 indicates pegs of adsorption free energy and adsorption equilibrium constant of weightloss experiment of diverse temperature regime. The adsorptive energy of adsorption is negative and the value is lower than 20 kJ mol^{-1} which indicates that the adsorption of the inhibitor on the metal surface was spontaneous, and that the adsorption was based on

physisorption mechanism (Vaszcilesin 2023; Noor 2008; Oguzie 2011 ; Martinez 2002). Equilibrium adsorption constant does not vary with temperature, and hence it means the desorption of inhibitor molecule cannot be induced much and therefore the inhibitor is also able to prevent the corrosion of metal surfaces at a high temperature.

Table.2 Free energy of adsorption ($\Delta G^\circ_{\text{ads}}$) and adsorption equilibrium constant (K)

Temperature (K)	$\Delta G^\circ_{\text{ads}}$ (kJ mol ⁻¹)	Adsorption equilibrium constant (K)
308	-10.0798	1.76×10^{-2}
318	-10.8303	1.76×10^{-2}
323	-11.9869	1.76×10^{-2}

It can be observed that the corrosion inhibition efficiency increased with temperature which is in agreement with the above adsorption isotherm value. The fact that the efficiency of the inhibitors increases as a result of the temperature suggests that the attractive force between the

Phenyl-o-Thiourea and metal surface is not diminished. Thus, an intense physisorption of the inhibitor and Al - metal surface exists.

Table.3 Effect of Temperature on Corrosion Rate(Inhibitor: Phenyl-o-thiourea)*

Temperature (K)	Weight loss (g)	Corrosion rate (mpy)	Inhibitor efficiency (%)
308	0.0410	0.2291	48
318	0.0351	0.1965	52
323	0.0310	0.1733	61

Table.4 gives the values of energy of activation (E_a) at different temperature regimes. Based on this tables it is evident that energy of activation also rises with

temperature and this denotes that rate of corrosion also falls with temperature.

Table.4 E_a at various temperature ranges

Temperature range (K)	Activation energy, E_a (kJ mol ⁻¹)
308 – 318	10.730
318 – 323	12.489
308 – 323	15.538

3.2 Electrochemical Impedance Spectroscopy

Figure. 4 represents the experimental Nyquist plots of the corrosion of Al in alkaline medium in the presence of the Phenyl-o- Thiourea. The electrochemical results obtained using faradic impedance spectroscopic

analysis could be accepted through the interpretation of the corrosion reaction as an equivalent circuit as shown in the Figure 5. Table.5 shows the parameters of impedance calculated based on the equivalent circuit.

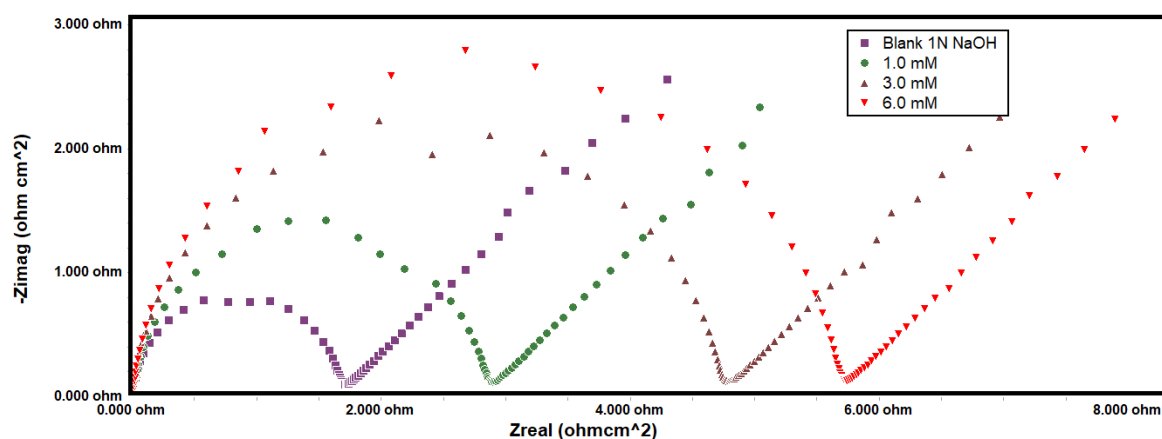


Figure.4 Nyquist plots between Al corrosion in alkaline medium with and without Phenyl-o-Thiourea.

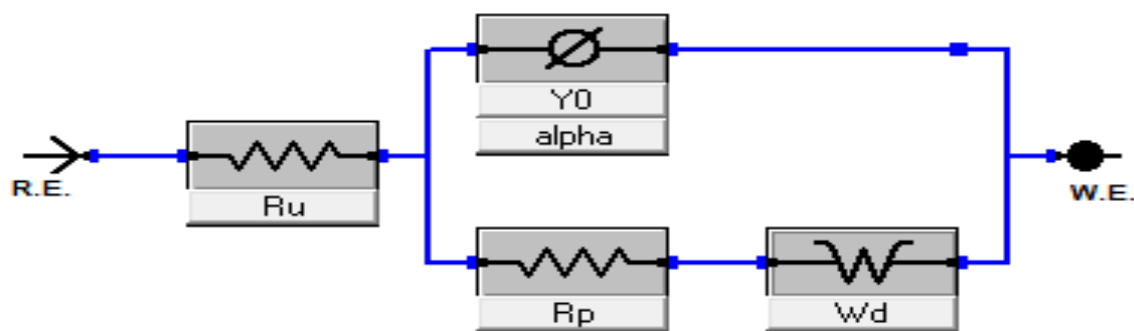


Fig.5: An equivalent circuit is used in extracting Electrochemical Impedance parameters of experiment Nyquist plots.

Table.5 Electrochemical Impedance Spectroscopic Parameters for Phenyl-o-thiourea Monomer on Aluminium Electrode in 1N NaOH

Inhibitor concentration (mM)	R_t ($\Omega \text{ m}^2$)	C_{dl} (F cm^2)
Blank	1.6368	7.0074×10^{-5}
1.0	2.2115	3.4805×10^{-5}
3.0	4.3393	2.1378×10^{-5}
6.0	1.6869	6.6440×10^{-5}

Evaluation of the results in this table shows that these impedance data can be fitted to best semicircles. It is also clear that the impounded reactions of aluminium in free solutions are greatly disturbed upon the introduction of Phenyl-o-thiourea which finally results in the height of R_{ct} values with the escalation of the thickness of such inhibitors. Also, it is indicated that the higher the concentration, the higher is the R_{ct} values of the inhibitors implying that the greater the concentration of the inhibitors the greater the inhibition efficiency. Reduction in the C_{dl} values with increase in concentration of inhibitors also lends support to the interpretation which was made regarding the weight loss data of inhibition by adsorption mechanism. This is owing to the fact that, whenever a greater adsorption falls on metal surface the C_{dl} value consistently reduces (Abbas 2022; Mobin 2023 ; Solomon 2021; Pushpanjali, 2017)[23-26].

3.3 Potentiodynamic Polarization Method

The potentiodynamic analysis has been performed on the aluminium in 1N NaOH with and without an inhibitor. Fig. 6 illustrates the curves of the polarization studies of the aluminium in the presence of 1N NaOH with and without the inhibitor. Anodic and cathodic were extrapolated through the provision of the electrochemical

parameters of E_{corr} (corrosion potential) and I_{corr} (corrosion current density). The Tafel constants β_a and β_c were also established with the help of the anodic and the cathodic polarization curves respectively. These parameters were given in table 6.

Figure.6 Tafel plots for corrosion of Al in 1N NaOH with and without inhibitor

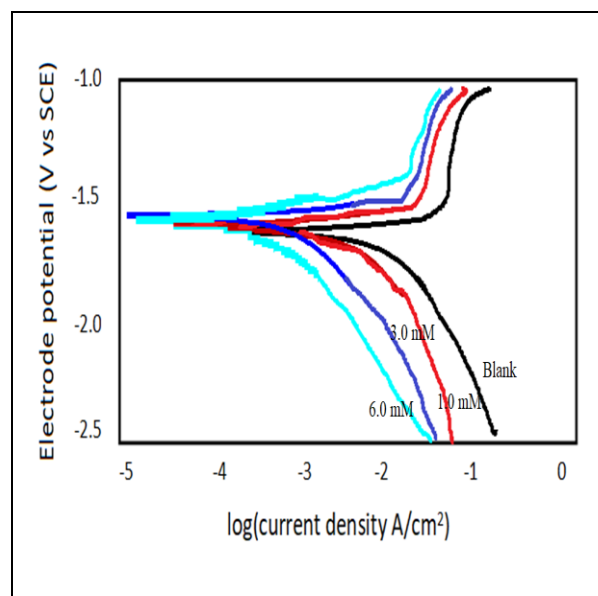


Table 6 Potentiodynamic Polarization curves of Corrosion of Al in 1N NaOH with and without Phenyl-o-thiourea.

Inhibitor concentration (mM)	β_c (mV)	β_a (mV)	E_{corr} (V)	I_{corr} (A m^{-2})	Corrosion rate (mmpy)
Blank	155.5	349.6	-1.6426	0.01347	5791.5
1.0	136.89	111.9	-1.6296	0.002409	1035.6
1.5	129.02	109.03	-1.6518	0.00187	804.6
3.0	131.58	55.547	-1.6681	0.000138	59.546

It is apparent that the addition of the inhibitor has impeded on the attack of the corrosive environment on aluminium. Additionally, as the concentration of the inhibitors increases, the current density of anodic and cathodic current also decreases which implies that the type of the inhibitors employed is mixed type inhibitor (Gece *et al.* 2008; Mansfeld *et al.*, 1990).

4. CONCLUSION

The following were the conclusions of the current research.

As the concentration increased, the inhibitor efficiency of the inhibitor increased. Temperature can have a very strong impact on the rate of corrosion and inhibition effectiveness. The adsorbent surfaces of metal surface adopt Langmuir adsorption isotherm. Arrhenius parameters include the energy of activation (E_a) and the change on free energy (ΔG) of adsorption which have suggested that physisorption was preceded by the adsorption of the inhibitor to Al surface in 1N NaOH. The study by potentiodynamic research polarization on the corrosion of the Aluminium and its inhibition with the use of the Phenyl-o-thiourea establishes that it is a mixed-type of an inhibitor.

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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REFERENCES

- Abbas MA, Ismail AS, Zakaria K, El-Shamy AM, El Abedin SZ. Adsorption, thermodynamic, and quantum chemical investigations of an ionic liquid that inhibits corrosion of carbon steel in chloride solutions. *Sci. Rep.*, 12(1), 12536 (2022).
<https://doi.org/10.1038/s41598-022-16534-3>
- Ammar, H. R., Moreau, C., Samuel, A. M., Samuel, F. H., & Doty, H. W.. Influences of alloying elements, solution treatment time and quenching media on quality indices of 413-type Al-Si casting alloys. *Materials Science and Engineering: A*, 489(1-2), 426-438, (2008).
<https://doi.org/10.1016/j.msea.2007.12.032>
- Bharatiya, U., Gal, P., Agrawal, A., Shah, M., & Sircar, A. Effect of corrosion on crude oil and natural gas pipeline with emphasis on prevention by ecofriendly corrosion inhibitors: a comprehensive review. *Journal of Bio-and Tribo-Corrosion*, 5(2), 35, (2019).
<https://doi.org/10.1007/s40735-019-0225-9>
- Gece G. The use of quantum chemical methods in corrosion inhibitor studies. *Corros. Sci.*, 50(11), 2981-2992 (2008).
<https://doi.org/10.1016/j.corsci.2008.08.043>
- Jenkins SJ. Aromatic adsorption on metals via first-principles density functional theory. *Proc. R. Soc. A*, 465(2110), 2949-2976 (2009).
<https://doi.org/10.1098/rspa.2009.0211>
- Liu W, Tkatchenko A, Scheffler M. Modeling adsorption and reactions of organic molecules at metal surfaces. *Acc. Chem. Res.*, 47(11), 3369-3377 (2014).
<https://doi.org/10.1021/ar500190p>
- Mahdi BS, Habeeb HM, Aziz IA, Hanoon MM, Sayyid FF, Mustafa AM, Gaaz TS, Jaaz AH, Khadom AA, Yousif E, Alamiery A. Understanding the impact of inhibitor concentration, immersion periods, and temperature on the corrosion inhibition of 2-piperazin-1-yl-1,3-benzothiazole in HCl solution. *Int. J. Corros. Scale Inhib.*, 13(2), 1164-1185 (2024).
- Mallaiya K, Subramaniam R, Srikandan SS, Gowri S, Rajasekaran N, Selvaraj A. Electrochemical characterization of the protective film formed by the unsymmetrical Schiff's base on the mild steel surface in acid media. *Electrochim. Acta*, 56(11), 3857-3863 (2011).
<https://doi.org/10.1016/j.electacta.2011.01.011>
- Mansfeld F. Electrochemical impedance spectroscopy (EIS) as a new tool for investigating methods of corrosion protection. *Electrochim. Acta*, 35(10), 1533-1544 (1990).
[https://doi.org/10.1016/0013-4686\(90\)80007-B](https://doi.org/10.1016/0013-4686(90)80007-B)
- Martinez S, Stern I. Thermodynamic characterization of metal dissolution and inhibitor adsorption processes in the low carbon steel/mimosa tannin/sulfuric acid system. *Appl. Surf. Sci.*, 199(1-4), 83-89 (2002).
[https://doi.org/10.1016/S0169-4332\(02\)00546-9](https://doi.org/10.1016/S0169-4332(02)00546-9)
- Mobin M, Zamindar S, Banerjee P. Mechanistic insight into adsorption and anti-corrosion capability of a novel surfactant-derived ionic liquid for mild steel in HCl medium. *J. Mol. Liq.*, 385, 122403 (2023).
<https://doi.org/10.1016/j.molliq.2023.122403>
- Mungwari CP, Obadele BA, King'onde CK. Synergistic corrosion inhibition of 304 stainless steel in 1 M HCl by methanolic *Acacia mearnsii* bark extract and KI: Thermodynamic, kinetic, and thermal analysis. *Chem. Thermodyn. Therm. Anal.*, 100233 (2025).
- Netzer FP, Ramsey MG. Structure and orientation of organic molecules on metal surfaces. *Crit. Rev. Solid State Mater. Sci.*, 17(5), 397-475 (1992).
<https://doi.org/10.1080/10408439208244599>
- Noor EA, Al-Moubaraki AH. Thermodynamic study of metal corrosion and inhibitor adsorption processes in mild steel/1-methyl-4[4'-(X)-styryl pyridinium iodides]/hydrochloric acid systems. *Mater. Chem. Phys.*, 110(1), 145-154 (2008).
<https://doi.org/10.1016/j.matchemphys.2008.01.033>

- Oguzie EE, Li Y, Wang SG, Wang F. Understanding corrosion inhibition mechanisms—Experimental and theoretical approach. *RSC Adv*, 1(5), 866–873 (2011).
<https://doi.org/10.1039/C1RA00065A>
- Rayung, M., Aung, M. M., Azhar, S. C., Abdullah, L. C., Su'ait, M. S., Ahmad, A., & Jamil, S. N. A. M. Bio-based polymer electrolytes for electrochemical devices: Insight into the ionic conductivity performance. *Materials*, 13(4), 838, (2020).
<https://doi.org/10.3390/ma13040838>
- Pan B, Xing B. Adsorption mechanisms of organic chemicals on carbon nanotubes. *Environ. Sci. Technol.*, 42(24), 9005–9013 (2008).
<https://doi.org/10.1021/es801777n>
- Pushpanjali, Rao SA, Rao P. Corrosion inhibition and adsorption behavior of *Murraya koenigii* extract for corrosion control of aluminum in hydrochloric acid medium. *Surf. Eng. Appl. Electrochem.*, 53(5), 475–485 (2017).
<https://doi.org/10.3103/S1068375517050114>
- Raja PB, Sethuraman MG. Natural products as corrosion inhibitors for metals in corrosive media A review. *Mater. Lett.*, 62(1), 113–116 (2008).
<https://doi.org/10.1016/j.matlet.2007.04.079>
- Sherif ESM, Park SM. Effects of 1,4-naphthoquinone on aluminum corrosion in 0.50 M sodium chloride solutions. *Electrochim. Acta*, 51(7), 1313–1321 (2006).
<https://doi.org/10.1016/j.electacta.2005.06.018>
- Solomon MM, Onyeachu IB, Njoku DI, Nwanonenyi SC, Oguzie EE. Adsorption and corrosion inhibition characteristics of 2-(chloromethyl) benzimidazole for C1018 carbon steel in a typical sweet corrosion environment: Effect of chloride ion concentration and temperature. *Colloids Surf. A Physicochem. Eng. Asp.*, 610, 125638 (2021).
<https://doi.org/10.1016/j.colsurfa.2020.125638>
- Vaszilcsin CG, Putz MV, Kellenberger A, Dan ML. On the evaluation of metal-corrosion inhibitor interactions by adsorption isotherms. *J. Mol. Struct.*, 1286, 135643 (2023).
<https://doi.org/10.1016/j.molstruc.2023.135643>
- Yang K, Xing B. Adsorption of organic compounds by carbon nanomaterials in aqueous phase: Polanyi theory and its application. *Chem. Rev.*, 110(10), 5989–6008 (2010).
<https://doi.org/10.1021/cr100059s>
- Zaera F. An organometallic guide to the chemistry of hydrocarbon moieties on transition metal surfaces. *Chem. Rev.*, 95(8), 2651–2693 (1995).
<https://doi.org/10.1021/cr00040a012>